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2 **(a)** A satellite is in a circular orbit around the Earth.

(i) State the direction of the force acting on the satellite.

..... [1]

(ii) A rocket is used to carry the satellite into space.

The initial mass of the rocket and satellite is 1.5×10^4 kg.

The initial force provided by the rocket is 1.9×10^5 N.

Calculate the initial acceleration of the rocket.

acceleration = ... m/s^2 [3]

(b) A large balloon filled with helium floats in the Earth's atmosphere.

(i) Explain, in terms of particles, how the helium gas exerts a pressure on the inside of the balloon.

.....

.....

..... [3]

(ii) The temperature of the air at the balloon's highest point above the Earth is 228 K.

Determine, in °C, the temperature of the air at this height.

temperature = ...°C [1]

[Total: 8]

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